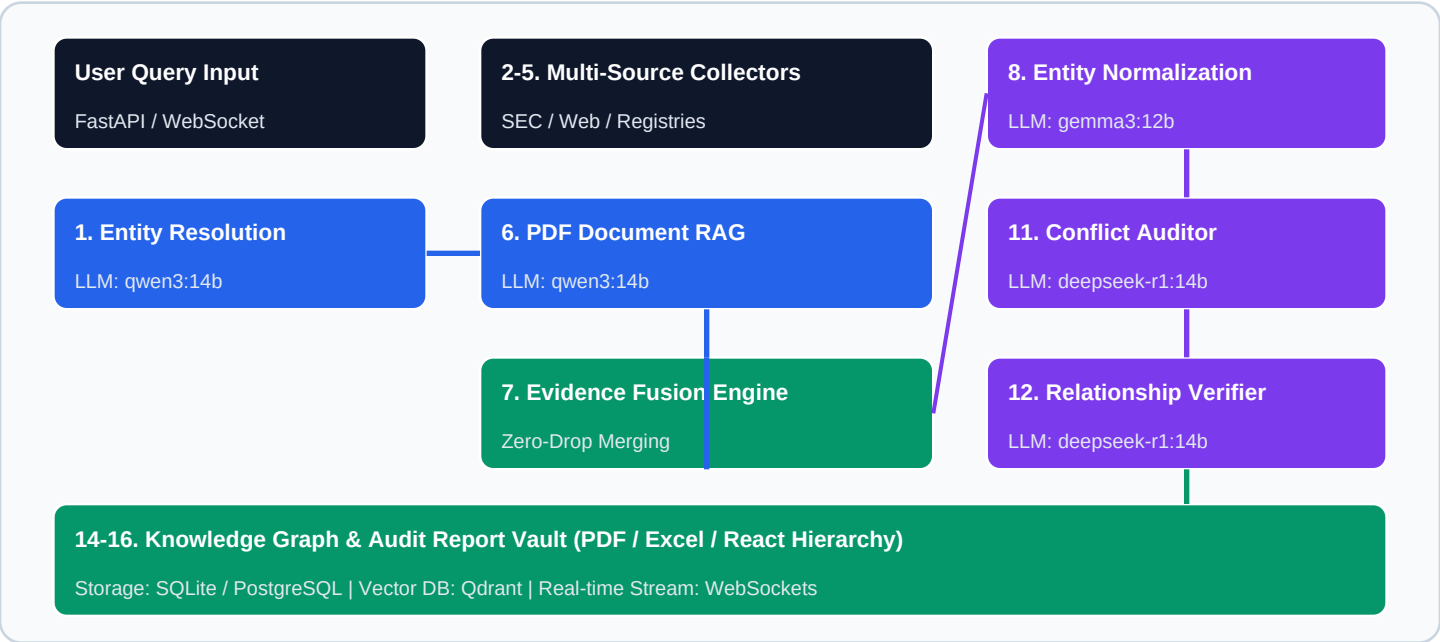


Corporate Subsidiary Intelligence Platform

Full Multi-Agent & LLM Architecture Plan

Executive Summary: An enterprise-grade multi-agent RAG platform designed to audit, discover, normalize, verify, and map global corporate subsidiary hierarchies. The engine executes live multi-source retrieval across SEC EDGAR filings, public registries, web pages, and annual reports to produce interactive hierarchy graphs and compliant PDF/Excel audit packages.

1. System Architecture Diagram (With LLM Placement)



2. LLM Placement & Multi-Model Mapping

The platform uses a **specialized multi-model architecture** routing tasks to distinct local open-weights LLMs via our Multi-Model Capability Router:

Pipeline Node	LLM Model Assigned	Task Capability & Role	Fallback Chain
entity_resolution	qwen3:14b	Converts fuzzy query strings into legal entity names, CIKs, tickers, and domains.	qwen2.5:14b → deepseek-r1:8b
document_intelligence	qwen3:14b	Contextual chunking & Q&A; over PDF annual reports in Qdrant vector DB.	qwen2.5:14b → deepseek-r1:8b
entity_normalization	gemma3:12b	Standardizes spelling variants and corporate suffixes (LLC, GmbH, Inc., Ltd.).	gemma2:9b → qwen2.5:14b
conflict_resolution	deepseek-r1:14b	Deep Reasoning: Audits contradictory ownership claims (divested vs owned) via reasoning chains.	deepseek-r1:8b → qwen2.5:14b
relationship_verification	deepseek-r1:14b	Deep Verification: Validates parent-child relationships against extracted evidence context.	deepseek-r1:8b → qwen2.5:14b

report_agent	qwen3:14b	JSON & Text Generation: Synthesizes narrative executive summaries for PDF/Excel audit packages.	qwen2.5:14b ➔ gemma2:9b
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3. Specialized Document Parsing Suite Included

Task / File Type	Integrated Library	Purpose & Role in Pipeline
SEC EDGAR Downloader	sec-edgar-downloader	Downloads automated 10-K, 10-Q, and Exhibit 21 filing attachments locally.
SEC EDGAR Search & Metadata	edgartools	Python interface for SEC EDGAR company metadata, CIK lookups, and filing text.
SEC Document Tree Parsing	sec-parser	Parses SEC HTML filings into structured document trees (sections, headings, tables).
HTML SEC Filings Parsing	BeautifulSoup + lxml	High-performance parsing of raw HTML/XML SEC filing exhibits.
XBRL / Financial Data	Arelle	Standard engine for reading XBRL and XML financial statements.
PDF Text & Layout Extraction	pdfplumber + pypdf	Extracts text blocks and layout geometries from corporate annual report PDFs.
PDF Subsidiary Table Extraction	Camelot	Converts tabular PDF subsidiary lists into structured pandas DataFrames.

4. Key Architectural Guarantees

- **1. Zero-Drop Data Retention Policy:** Every candidate entity discovered by collectors is preserved in the output tree. Unconfirmed items are marked as 'Unverified' rather than deleted.
- **2. 100% Local Inference Execution:** Fully operational offline using local Ollama LLMs (qwen2.5:14b & deepseek-r1:8b) with zero external API dependencies required.
- **3. Multi-Factor Dynamic Confidence Scoring:** Scores entities from 0% to 100% based on regulatory authority (SEC EDGAR: 50%, Public Registries: 40%, Web: 10%) plus cross-validation bonuses.
- **4. Real-Time Telemetry & WebSockets:** Live streaming of node execution timings, entity count metrics, and candidate names to the React/Vite dashboard.